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Lehrbuch der Algebra

Working English Translation Batch 06

Reduction of Abelian Equations and the Beginning of Lagrange Resolvents

Translator's note

This working English translation continues the current Weber lane from the cleaned source in hand. It covers the transition from Abelian equations to cyclic equations and the opening of Weber's treatment of Lagrange resolvents. I have preserved Weber's order of argument and formulas while normalizing punctuation and a few typographical conventions.

§4. Cyclic equations

163. Reduction of Abelian equations to cyclic equations

We have already seen above that in a transitive Abelian group, besides the identity, no permutation occurs that leaves any symbol fixed. From this one obtains another theorem.

Assume that a permutation π of such a group is decomposed into its cycles, and let the cycle with the fewest elements be an r -cycle. Then π^r leaves the elements of this cycle fixed and therefore must be the identity permutation. But from $\pi^r = 1$ it follows that, if other cycles are present, they too must have r elements. Thus:

Theorem. A permutation of a transitive Abelian group contains only cycles of the same length.

If m is the degree of the group, then the number r of elements in one cycle must divide m , and if $m = rs$, then s is the number of r -cycles of which π consists.

Now let P be the group of an Abelian equation $F(x) = 0$, and choose any non-identity permutation π in P . Decompose it into cycles

$$\pi = \gamma_1 \gamma_2 \cdots \gamma_s, \tag{1}$$

where each cycle γ refers to r roots of the equation $F(x) = 0$. Arrange these roots so that

$$\begin{aligned} \gamma_1 &= (\alpha, \alpha_1, \alpha_2, \dots, \alpha_{r-1}), \\ \gamma_2 &= (\beta, \beta_1, \beta_2, \dots, \beta_{r-1}), \\ &\dots, \\ \gamma_s &= (\delta, \delta_1, \delta_2, \dots, \delta_{r-1}). \end{aligned} \tag{2}$$

If π_i is any permutation of P , then because of commutativity one has

$$\pi^{-1} \pi_i \pi = \pi_i, \quad \text{that is,} \quad \pi_i \pi = \pi \pi_i. \tag{3}$$

By the theorem on the formation of the permutations $\pi^{-1}\pi_i\pi$ (§154, no. 6), it follows that γ_1 cannot be changed when the permutations π are carried out. But since γ_1 is completely determined inside the cycles of π , apart from its ordering and its initial element, it follows that under any permutation π_i in P the elements belonging to one cycle γ cannot be separated from one another; they can only be cyclically permuted among themselves, and the cycles can moreover be permuted among one another.

Hence, if $s > 1$, the group is imprimitive.

A rational function of the arguments $\alpha, \alpha_1, \dots, \alpha_{r-1}$ that does not change its value when the α are subjected to the cyclic permutation γ_1 and its repetitions is called a *cyclic function* of the α . If we denote by $\omega, \omega_1, \dots, \omega_{s-1}$ the conjugate values of ω , setting for instance

$$\begin{aligned}\omega &= \psi(\alpha, \alpha_1, \dots, \alpha_{r-1}), \\ \omega_1 &= \psi(\beta, \beta_1, \dots, \beta_{r-1}), \quad \dots,\end{aligned}\tag{4}$$

then these quantities are, by §158, the roots of an irreducible equation of degree s ,

$$\Phi(\omega) = 0,\tag{5}$$

whose group is obtained by determining the permutations induced on the quantities (4) by the permutations coming from P .

But the permutation of the quantities (4) induced by $\pi_i\pi_j$ is obtained by composing the permutations induced separately by π_i and π_j ; hence the group of permutations of (4) is itself commutative.

Therefore $\Phi(\omega) = 0$ is an Abelian equation of degree s . Its first factor $F(x, \omega)$ has the roots $\alpha, \alpha_1, \dots, \alpha_{r-1}$, and the group of the equation

$$F(x, \omega) = 0$$

consists only of the permutations belonging to the period of the cyclic permutation γ_1 .

We shall call an equation whose group consists of a single cycle and its repetitions a *cyclic equation*. Thus cyclic equations are the simplest special case of Abelian equations.

We have therefore proved that the solution of every Abelian equation is reduced to the solution of an Abelian equation of lower degree together with the solution of a collection of cyclic equations.

One can apply the same theorem again to the auxiliary equation of degree s , and continue in this way until that auxiliary equation is reduced to degree one.

Thus we obtain the result:

Theorem. The solution of an Abelian equation can always be reduced to the solution of a sequence of cyclic equations whose degrees divide the degree of the given equation.

It is not necessary to include irreducibility in the definition of cyclic equations. We may therefore formulate the definition in general as follows.

An equation of degree m , $F(x) = 0$, with m distinct roots is called a *cyclic equation* over the field K if its roots $\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}$ are not rational, but can be arranged in such a way that the cyclic functions of the roots are rational over K .

If therefore

$$\pi = (\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}),\tag{6}$$

then every function belonging to K and invariant under the permutations of the cyclic group

$$C = 1, \pi, \pi^2, \pi^3, \dots, \pi^{m-1}\tag{7}$$

must be rational.

The Galois group of a cyclic equation is either the whole period C , in which case the equation is irreducible, or else it is a subgroup of C ; in the latter case, if e and f are two integer divisors of m with $m = ef$, then this group consists of the permutations

$$C^e = 1, \pi^e, \pi^{2e}, \dots, \pi^{(f-1)e}, \quad (8)$$

and the cyclic equation $F(x) = 0$ splits into e factors of degree f .

For let

$$t = \psi(\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e})$$

be a function belonging to the group C , and assume it is equal to a quantity in K , which we denote by a .

Then on the rational equation

$$F(x) = F(x, a)F(x, a_1) \cdots F(x, a_{e-1})$$

no permutation not contained in C can act, and consequently the group of the equation can contain no substitutions other than those occurring in C .

If now π itself occurs in the group of the equation, then the group is identical with C , and since C is transitive, the equation is irreducible.

If, however, $\pi^{e'}$ is the smallest power of π occurring in the group of the equation, then $C^{e'}$ is the group. If $e' < m$, then $C^{e'}$ is intransitive, and the equation is reducible.

The same reduction can also be applied to cyclic equations whose degree is not prime. If namely $m = ef$ is any factorization of m into two factors, then the permutation π^e splits into e cycles γ , each with f elements:

$$\gamma = (\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e}). \quad (9)$$

Now determine a function ψ belonging to the first of these cycles, and set

$$\begin{aligned} \eta &= \eta_0 = \psi(\alpha, \alpha^e, \alpha^{2e}, \dots, \alpha^{(f-1)e}), \\ \eta_1 &= \psi(\alpha_1, \alpha^{e+1}, \alpha^{2e+1}, \dots, \alpha^{(f-1)e+1}), \\ &\dots, \\ \eta_{e-1} &= \psi(\alpha_{e-1}, \alpha_{2e-1}, \alpha_{3e-1}, \dots, \alpha_{m-1}). \end{aligned} \quad (10)$$

These quantities are all distinct, yet under application of the permutation π they pass cyclically into one another. Hence their cyclic functions, and therefore also their symmetric functions, are rational. So they are the roots of a cyclic equation of degree e , while $F(x)$ splits into e factors of degree f ,

$$F(x) = F(x, \eta)F(x, \eta_1) \cdots F(x, \eta_{e-1}),$$

each of which gives a cyclic equation of degree f for the roots of one of the cycles γ .

Indeed, if for example

$$F_1 = (x - \alpha)(x - \alpha^e) \cdots (x - \alpha^{(f-1)e}),$$

then this function admits the permutations in the period of γ . It also admits the permutations of the group that becomes the Galois group of our equation after adjoining η , a group that can consist only of powers of $\gamma_0, \gamma_1, \dots, \gamma_{e-1}$.

Hence, by Lagrange's theorem (§155, equation (2)), F_1 is rationally expressible in terms of η , so $F_1 = F(x, \eta)$. But the equation

$$F(x, \eta) = 0$$

is again cyclic over $K(\eta)$, since the cyclic functions of its roots belong to that field.

Like all Abelian equations, cyclic equations have the property that every root is rationally expressible in terms of every other. Here these expressions can be arranged in the following cyclic manner. If $\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}$ are the roots of the cyclic equation $F(x) = 0$, then the entire function of degree $m - 1$ in x

$$\Psi_m\left(x + \frac{F'(x)}{\Psi'_m(x)}\right) = \Psi_m$$

is unchanged by the permutation π and therefore belongs to K . If in it we set

$$x = \alpha, \alpha_1, \dots, \alpha_{m-1}$$

and introduce the symbol θ , then it follows that

$$\alpha_1 = \theta(\alpha), \quad \alpha_2 = \theta(\alpha_1), \quad \dots, \quad \alpha_{m-1} = \theta(\alpha_{m-2}), \quad \alpha = \theta(\alpha_{m-1}), \quad (11)$$

and this holds whether $F(x)$ is reducible or irreducible, provided only that $F(x)$ and $F'(x)$ have no common factor.

The solution of cyclic equations of degree m has thus been made dependent on the solution of cyclic equations whose degrees are the prime factors of m . If for example m is a power of 2, then the solution is carried out by a sequence of square roots. In this way Gauss first treated the cyclotomic equations.

Finally we add the remark, important for what follows, that for prime degree the notions of normal equation and cyclic equation coincide. Certainly a cyclic equation of prime degree is irreducible and hence a normal equation.

Conversely, if the degree n of a normal equation, which is also the degree of its group, is prime, and if π is any non-identity permutation of this group, then the degree of π , which must divide n , is equal to n ; hence

$$\pi^n = 1, \quad \pi, \pi^2, \dots, \pi^{n-1}$$

constitute the group of the equation, and the equation is cyclic.

§5. Lagrange resolvents

164. The method of solution for cyclic equations

The method by which we shall now learn to solve cyclic equations applies uniformly to prime degrees and to composite degrees. One uses for this purpose certain expressions known under the name of *Lagrange resolvents*, which are of great use in all investigations concerning the algebraic solution of equations.

Let $F(x) = 0$ be an equation with roots

$$\alpha, \alpha_1, \alpha_2, \dots, \alpha_{m-1}.$$

Let ε be any m th root of unity, and introduce the notation

$$(\varepsilon, \alpha) = \alpha + \varepsilon\alpha_1 + \varepsilon^2\alpha_2 + \dots + \varepsilon^{m-1}\alpha_{m-1}. \quad (1)$$

These sums are what one calls the *Lagrange resolvents*. If these functions are known for all m th roots of unity ε , then the equation itself is solved; for by §133, equation (6),

$$\sum_{\varepsilon} \varepsilon^{-k} = m \quad \text{or} \quad 0, \quad (2)$$

according as k is divisible by m or not, and hence

$$m\alpha_k = \sum_{\varepsilon} (\varepsilon, \alpha) \varepsilon^{-k}, \quad (3)$$

where the sums extend over all m th roots of unity ε .

The other roots can be expressed in the same way:

$$m\alpha_k = \sum_{\varepsilon} \varepsilon^{-k}(\varepsilon, \alpha), \quad (4)$$

so that everything is indeed reduced to knowledge of (ε, α) and the functions formed from it.

The sums (2) and (3) can also be written in another fashion, since all m th roots of unity are powers of a primitive one.

If therefore ε denotes a fixed primitive m th root of unity and h runs through a complete system of residues modulo m , say $0, 1, \dots, m-1$, then equations (2), (3) may be written as

$$m\alpha = \sum_h (\varepsilon^h, \alpha), \quad m\alpha_k = \sum_h \varepsilon^{-hk}(\varepsilon^h, \alpha). \quad (5)$$

We first investigate these resolvents as functions of m independent variables α , and for simpler notation we adopt the convention that $\alpha_h = \alpha_k$ whenever $h \equiv k \pmod{m}$.

Then the entire system of variables α is obtained by letting the index h run through a complete residue system modulo m .

For these resolvents the following statements hold.

1. If one applies to the indices of the α the cyclic permutation

$$\pi = (0, 1, \dots, m-1),$$

then (ε, α) passes into $\varepsilon^{-1}(\varepsilon, \alpha)$, and under the permutation π^k it passes into $\varepsilon^{-k}(\varepsilon, \alpha)$.

This follows immediately from the defining formula (1).

Now let ν be any positive exponent and form $(\varepsilon, \alpha)^\nu$. Expand this power by the polynomial theorem, set $\varepsilon^m = 1$, and order the result by powers of ε . One obtains an expression of the form

$$(\varepsilon, \alpha)^\nu = A_0^{(\nu)} + \varepsilon A_1^{(\nu)} + \varepsilon^2 A_2^{(\nu)} + \dots + \varepsilon^{m-1} A_{m-1}^{(\nu)}, \quad (5')$$

where the $A_h^{(\nu)}$ are forms of degree ν with integer coefficients in the variables α , but are independent of ε . Again we stipulate that $A_h^{(\nu)} = A_k^{(\nu)}$ whenever $h \equiv k \pmod{m}$.

From this we prove the following statement.

2. If one applies the cyclic permutation π to the indices of the α , then the indices of the coefficients in $(\varepsilon, \alpha)^\nu$ undergo the cyclic permutation π^ν ; that is, $A_h^{(\nu)}$ passes into $A_{h+\nu}^{(\nu)}$.

To prove this, note that formula (5') remains valid for every m th root of unity ε , including 1. Using (2) one then obtains

$$mA_k^{(\nu)} = \sum_{\varepsilon} \varepsilon^{-k}(\varepsilon, \alpha)^\nu. \quad (6)$$

If we apply the permutation π on the right-hand side, then by statement 1 the term $(\varepsilon, \alpha)^\nu$ is multiplied by $\varepsilon^{-\nu}$, so the whole sum becomes

$$\sum_{\varepsilon} \varepsilon^{-(k+\nu)}(\varepsilon, \alpha)^\nu = mA_{k+\nu}^{(\nu)},$$

which is exactly the assertion.

3. Thus, when the permutation π is applied to the indices of the α , it induces on the indices of the coefficients of the product ordered by powers of ε the permutation determined by the total exponent-weight.

This general principle will be used in the further computation of the Lagrange resolvents.

This batch stops at the opening structural properties of the Lagrange resolvents.